

Part A: Negative Number Operations*Evaluate each expression completely.*

1. $3 - 37$

11. $37 + 11 \cdot 15$

2. $17 - 8$

12. $17 + 14 \cdot (-5)$

3. $25 + 19 + (-17)$

13. $-22 + 6 \cdot 4$

4. $-7 - (-15) + (-24)$

14. $-20 + (-8) \cdot 14$

5. $-10 + (-38) + 13$

15. $-156 \div (-13) - (-8)$

6. $-7 \cdot 5$

16. $3^2 - (-11) \cdot 8$

7. $-8 \cdot (-7)$

17. $(-9)^3 - 15 \cdot 15$

8. $-78 \div (-13)$

18. $21 - 14 \cdot (5 + 21)$

9. $14 \cdot (-9)$

19. $-27 - 8 \cdot ((-9) + (-27))$

10. $15 \cdot (-15)$

20. $18 - 11 \cdot ((-5) + 18)$

Part B: Fractions*Evaluate. Write every answer in lowest terms.*

1. $\frac{2}{3} - \frac{1}{2}$

3. $\frac{1}{6} + \frac{2}{3}$

2. $\frac{1}{2} - \frac{1}{3}$

4. $\frac{3}{5} - \frac{3}{4}$

5. $\frac{1}{5} - \frac{1}{3}$

11. $3\frac{1}{2} - 2\frac{1}{8}$

6. $\frac{4}{5} \div \frac{1}{3}$

12. $3\frac{2}{3} - 1\frac{1}{6}$

7. $\frac{1}{6} \div \frac{1}{5}$

13. $4\frac{2}{3} - 3\frac{5}{6}$

8. $\frac{3}{4} \cdot \frac{2}{3}$

14. $6 + 1\frac{4}{9}$

9. $\frac{2}{5} \div \frac{1}{2}$

15. $2\frac{2}{3} + 4\frac{1}{2}$

10. $\frac{1}{2} \cdot \frac{1}{2}$

Part C: Writing Fractions in Lowest Terms

Write each fraction in lowest terms.

1. $\frac{4}{32}$

4. $\frac{3}{30}$

2. $\frac{12}{8}$

5. $\frac{22}{8}$

3. $\frac{33}{12}$

Part D: Simplifying Expressions

Simplify each expression.

1. $-9x + 5 - 14x + 13$

2. $3x + 3y - 14 + 12x + 4y$

3. $19x + 7 - 16x + 18$

7. $2(2x + 8) + 11x + 11$

4. $-7x + 3 + 19x + 2$

8. $18(15x + 11) + 7x + 10$

5. $-3x + 7y + 18 + 13x + 7y$

9. $15(-10x + 20) + 16x + 2$

6. $15(17x + 7) + 17x + 13$

10. $19(2x + 17) + 10x + 11$

Part E: Adding and Subtracting Linear Expressions

Add or subtract, then simplify.

1. $(4x + 6) - (3x + 7)$

4. $(-15x + 10) + (13x - 16)$

2. $(3x - 7) - (-5x - 3)$

5. $(x + 13) + (-19x + 7)$

3. $(10x + 8) + (-19x + 14)$

Part F: Factoring Linear Expressions

Factor out the greatest common factor.

1. $48x + 18$

4. $9x + 81$

2. $50x + 70$

5. $11x - 66$

3. $5x + 30$

Part G: Solving Linear Equations

Solve each equation for x . Show your steps.

1. $x + 2 = -5$

2. $x - 2 = -11$

3. $6x = 54$

12. $3x - 5 = -x + 3$

4. $3x = 27$

13. $12x - 7 = -3x - 67$

5. $5x + 4 = 29$

14. $-12x - 9 = 10x - 31$

6. $4x - 9 = 19$

15. $\frac{x}{6} - 3 = -\frac{4}{3}$

7. $5x - 6 = -1$

16. $-\frac{5}{2}x - 16 = -\frac{27}{2}$

8. $5x - 9 = -4$

17. $-\frac{6}{5}x + 7 = \frac{89}{5}$

9. $5(x + 7) - 10 = 70$

18. $\frac{4x - 29}{4} = \frac{45}{12}$

10. $4(6x + 3) + 4 = 184$

19. $\frac{4x + 48}{4} = \frac{6x + 76}{8}$

11. $7(3x + 6) - 5 = -215$

20. $\frac{4x - 52}{10} = \frac{6x - 86}{5}$

Part H: Linear Inequalities*Solve each inequality for x .*

1. $-5x \geq -65$

11. $-16x - 11 \geq 7x + 81$

2. $-9x > -90$

12. $-8x - 12 \geq 19x - 93$

3. $-3x \leq -33$

13. $\frac{x}{5} - 14 < -\frac{83}{5}$

4. $x + 13 \geq 7$

14. $\frac{5}{6}x + 3 < 18$

5. $-10x - 6 < 14$

15. $\frac{x}{6} - 16 \leq -\frac{33}{2}$

6. $16x + 18 \leq 50$

16. $\frac{x-9}{5} + 9 > \frac{51}{5}$

7. $-5x + 10 > 50$

17. $\frac{7}{6}x - 16 > \frac{8}{5}x - \frac{311}{30}$

8. $-13x + 7 < 137$

18. $-2x + 11 < \frac{8}{5}x + \frac{127}{5}$

9. $-20x - 1 > 14x + 509$

10. $19x - 15 \geq 17x - 51$

19. $-\frac{7}{6}x - 3 \leq \frac{3}{2}x + \frac{31}{3}$

$$20. -\frac{6}{5}x - 3 \leq -\frac{3}{2}x + 3$$

Part I: Exponent Rules

Simplify. Write answers with positive exponents.

$$1. 3x^6 \cdot 2x^6$$

$$11. (5x^7)^4$$

$$2. 4x^4 \cdot 4x^6$$

$$12. 7 \cdot (6^1)^{-1}$$

$$3. x^5 \cdot x^2$$

$$13. \frac{5}{5^{-2}}$$

$$4. 3x^4 \cdot 4x^7$$

$$14. \frac{1}{3^{-3}}$$

$$5. \frac{3x^5}{x^3}$$

$$15. 5^3 \cdot 5^{-3}$$

$$6. \frac{x^8}{x^2}$$

$$16. 5^3 \cdot 5^0$$

$$7. \frac{2x^{10}}{x^5}$$

$$17. 6^5 \cdot 6^{-2}$$

$$8. \frac{4x^{10}}{x^3}$$

$$18. \frac{2x^{12}y^7}{2x^2y^2}$$

$$9. (8x^2)^2$$

$$19. (2x^{-2})^3 \cdot x^{12}$$

$$10. (7x^8)^4$$

$$20. \frac{9x^{12}y^4}{x^4y^3}$$

Answer Key

Part A: Negative Number Operations

- | | | |
|----------|------------|------------|
| 1. -34 | 8. 6 | 15. 20 |
| 2. 9 | 9. -126 | 16. 97 |
| 3. 27 | 10. -225 | 17. -954 |
| 4. -16 | 11. 202 | 18. -343 |
| 5. -35 | 12. -53 | 19. 261 |
| 6. -35 | 13. 2 | 20. -125 |
| 7. 56 | 14. -132 | |

Part B: Fractions

- | | | |
|--------------------|-------------------|--------------------|
| 1. $\frac{1}{6}$ | 6. $2\frac{2}{5}$ | 11. $1\frac{3}{8}$ |
| 2. $\frac{1}{6}$ | 7. $\frac{5}{6}$ | 12. $2\frac{1}{2}$ |
| 3. $\frac{5}{6}$ | 8. $\frac{1}{2}$ | 13. $\frac{5}{6}$ |
| 4. $-\frac{3}{20}$ | 9. $\frac{4}{5}$ | 14. $7\frac{4}{9}$ |
| 5. $-\frac{2}{15}$ | 10. $\frac{1}{4}$ | 15. $7\frac{1}{6}$ |

Part C: Writing Fractions in Lowest Terms

- | | | |
|-------------------|-------------------|-------------------|
| 1. $\frac{1}{8}$ | 3. $2\frac{3}{4}$ | 5. $2\frac{3}{4}$ |
| 2. $1\frac{1}{2}$ | 4. $\frac{1}{10}$ | |

Part D: Simplifying Expressions

- | | | |
|--------------------|---------------------|-----------------|
| 1. $18 - 23x$ | 5. $10x + 14y + 18$ | 9. $302 - 134x$ |
| 2. $15x + 7y - 14$ | 6. $272x + 118$ | 10. $48x + 334$ |
| 3. $3x + 25$ | 7. $15x + 27$ | |
| 4. $12x + 5$ | 8. $277x + 208$ | |

Part E: Adding and Subtracting Linear Expressions

- | | | |
|-------------|--------------|---------------|
| 1. $x - 1$ | 3. $22 - 9x$ | 5. $20 - 18x$ |
| 2. $8x - 4$ | 4. $-2x - 6$ | |

Part F: Factoring Linear Expressions

- | | | |
|-----------------|---------------|----------------|
| 1. $6(8x + 3)$ | 3. $5(x + 6)$ | 5. $11(x - 6)$ |
| 2. $10(5x + 7)$ | 4. $9(x + 9)$ | |

Part G: Solving Linear Equations

- | | | |
|-------------|------------|---------------|
| 1. $x = -7$ | 5. $x = 5$ | 9. $x = 9$ |
| 2. $x = -9$ | 6. $x = 7$ | 10. $x = 7$ |
| 3. $x = 9$ | 7. $x = 1$ | 11. $x = -12$ |
| 4. $x = 9$ | 8. $x = 1$ | 12. $x = 2$ |

13. $x = -4$

14. $x = 1$

15. $x = 10$

16. $x = -1$

17. $x = -9$

18. $x = 11$

19. $x = -10$

20. $x = 15$

Part H: Linear Inequalities

1. $x \leq 13$

2. $x < 10$

3. $x \geq 11$

4. $x \geq -6$

5. $x > -2$

6. $x \leq 2$

7. $x < -8$

8. $x > -10$

9. $x < -15$

10. $x \geq -18$

11. $x \leq -4$

12. $x \leq 3$

13. $x < -13$

14. $x < 18$

15. $x \leq -3$

16. $x > 15$

17. $x < -13$

18. $x > -4$

19. $x \geq -5$

20. $x \leq 20$

Part I: Exponent Rules

1. $6x^{12}$

2. $16x^{10}$

3. x^7

4. $12x^{11}$

5. $3x^2$

6. x^6

7. $2x^5$

8. $4x^7$

9. $64x^4$

10. $2401x^{32}$

11. $625x^{28}$

12. $\frac{7}{6}$

13. 125

14. 27

15. 1

16. 125

17. 216

18. $x^{10}y^5$

19. $8x^6$

20. $9x^8y$